2023BCSE008

Q1:Write a Java program to find the largest of three numbers.

Lab_03 > largestOfThreeNumbers.java > largestOfThreeNumbers > largest(int, int, int)

```java
package Lab_03;
import java.util.Scanner;
public class largestOfThreeNumbers {
    public static int largest(int a,int b,int c){
        if(a>b && a>c){
            return a;
        }else if(b>a && b>c){
            return b;
        }else{
            return c;
        }
    }
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter ist number:");
        int a = sc.nextInt();
        System.out.println(x: "Enter 2nd number:");
        int b = sc.nextInt();
        System.out.println(x: "Enter 3rd number:");
        int c = sc.nextInt();
        int result = largest(a, b, c);
        System.out.println("The laregst number is :"+result);
        sc.close();
```

OUTPUT:—

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac Lab_03 % java largestOfThreeNumbers.java
Enter ist number:
4
Enter 2nd number:
8
Enter 3rd number:
1
The laregst number is :8
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q2:Write a Java program to check whether a character is a vowel or consonant.

Lab_03 > vowelOrConsonant.java > vowelOrConsonant > main(String[])

~/Desktop/LabJava/Lab_03 • Contains emphasized items

```java
import java.util.Scanner;


public class vowelOrConsonant {
    Run | Debug
    public static void main(String[] args) {
     Scanner sc = new Scanner(System.in);
     System.out.println(x: "Enter the character:");
     char ch = sc.next().charAt(index: 0);
     if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U' ){

         System.out.println(x: "The Given Character is a vowel");
     }else{
         System.out.println(x: "The given character is a consonant");
     }
     sc.close();
    }

}
```

OUTPUT:——-

```
gauravmathur@Annus-Mac Lab_03 % java vowelOrConsonant.java
Enter the character:
A
The given character is a consonant
```

2023BCSE008

Q3:Write a Java program to calculate grade based on marks:

- 90+ → A
- 75–89 → B
- 50–74 → C
- Below 50 → Fail

Lab_03 > gradeCalculator.java > gradeCalculator > main(String[])

```java
/*Write a Java program to calculate grade based on marks:

    •   90+ → A
    •   75–89 → B
    •   50–74 → C
    •   Below 50 → Fail
*/
package Lab_03;
import java.util.Scanner;

public class gradeCalculator {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter Your marks:");
        int a = sc.nextInt();
        if(a>90){
            System.out.println(x: "Your grade is : A");
        }else if(a>75 || a<89){
            System.out.println(x: "Your grade is :B ");
        }else if(a>50 || a<74){
            System.out.println(x: "Your Grade is C ");
        }else{
            System.out.println(x: "You are Fail");
        }
        sc.close();
    }

}
```

OUTPUT:— —

```
gauravmathur@Annus-Mac Lab_03 % java gradeCalculator.java
Enter Your marks:
92
Your grade is : A
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q4:Write a Java program using switch to print day of the week based on number (1–7).

Lab_03 > Switch.java > Switch > main(String[])

```java
package Lab_03;
import java.util.Scanner;

public class Switch {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the number between (1-7) :");
        int a = sc.nextInt();
        switch(a){
            case 1:
                System.out.println(x: "Monday");
                break;
            case 2:
                System.out.println(x: "Tuesday");
                break;
            case 3:
                System.out.println(x: "Wednesday");
                break;
            case 4:
                System.out.println(x: "Thursday");
                break;
            case 5:
                System.out.println(x: "Friday");
                break;
            case 6:
                System.out.println(x: "Saturday");
                break;
            case 7:
                System.out.println(x: "Sunday");
                break;
            default:
                System.out.println(x: "Invalid Number");

        }
        sc.close();

    }

}
```

OUTPUT:—-

```
gauravmathur@Annus-Mac Lab_03 % javac Switch.java
gauravmathur@Annus-Mac Lab_03 % java Switch.java
Enter the number between (1-7) :
4
Thursday
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q5:Write a Java program to check whether a number is a multiple of 10 or not.

Lab_03 > multiple.java > multiple > main(String[])

```java
package Lab_03;
import java.util.Scanner;

public class multiple {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the number :");
        int a = sc.nextInt();
        if(a%10==0){
            System.out.println(a+" is a multiple of 10");
        }else{
            System.out.println(a+" is not a multiple of 10");
        }
        sc.close();
    }

}
```

OUTPUT:———

```
gauravmathur@Annus-Mac Lab_03 % java multiple.java
Enter the number :
20
20 is a multiple of 10
gauravmathur@Annus-Mac Lab_03 % java multiple.java
Enter the number :
44
44 is not a multiple of 10
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q6:Write a Java program to print numbers from 100 to 1 using a while loop.

Lab_03 > whilee.java > ...

```java
package Lab_03;

public class whilee {
    Run | Debug
    public static void main(String[] args) {
        int n = 100;
        while(n>=1){
            System.out.print(n+" ");
            n--;
        }
    }


}
```

OUTPUT:———-

```
gauravmathur@Annus-Mac Lab_03 % java whilee.java
100 99 98 97 96 95 94 93 92 91 90 89 88 87 86 85 84 83 82 81 80 79 78 77 76 75 74 73 72 71 70 69 68 67 66 65 64 63 62 61 60 59 58 57 56 55 54 53 52
 51 50 49 48 47 46 45 44 43 42 41 40 39 38 37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 %
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q7:Write a Java program to print the multiplication table of a given number.

Lab_03 > table.java > table > main(String[])

```java
package Lab_03;
import java.util.Scanner;

public class table {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter  number:");
        int a = sc.nextInt();
        for(int i=1;i<=10;i++){
            System.out.println(i +"x" +a+" is "+(i*a));
        }
        sc.close();
    }

}
```

OUTPUT:— — — — —-

```
gauravmathur@Annus-Mac Lab_03 % java table.java
Enter  number:
2
1x2 is 2
2x2 is 4
3x2 is 6
4x2 is 8
5x2 is 10
6x2 is 12
7x2 is 14
8x2 is 16
9x2 is 18
10x2 is 20
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q8:Write a Java program to calculate factorial of a number.

Lab_03 > factorial.java > ...

```java
package Lab_03;
import java.util.Scanner;

public class factorial {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter  number:");
        int a = sc.nextInt();
        int factorial= 1;
        for(int i=1;i<a;i++){
            factorial =factorial*i;
        }
        System.out.println("The Factorial is :"+ factorial);
        sc.close();

    }

}
```

Output:———

```
gauravmathur@Annus-Mac Lab_03 % javac factorial.java
gauravmathur@Annus-Mac Lab_03 % java factorial.java
Enter  number:
4
The Factorial is :6
```

2023BCSE008

Q9:Write a Java program to count number of digits in a number.

Lab_03 > digit.java > digit > main(String[])

```java
package Lab_03;

import java.util.Scanner;

public class digit {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter a number: ");
        int num = sc.nextInt();
        int count = 0;
        if (num == 0) {
            count = 1;
        } else {
            num = Math.abs(num);

            while (num > 0) {
                num = num / 10;
                count++;
            }
        }
        System.out.println("Number of digits = " + count);
        sc.close();
    }

}
```

OUTPUT:———

```
gauravmathur@Annus-Mac Lab_03 % java digit.java
Enter a number: 4532
Number of digits = 4
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q10:Write a Java program to reverse a number.

Lab_03 > reverse.java > ...

```java
package Lab_03;
import java.util.Scanner;

public class reverse {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter a number: ");
        int num = sc.nextInt();
        int reverse = 0;
        while (num != 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
        }
        System.out.println("Reversed number = " + reverse);
        sc.close();

    }
}
```

OUTPUT:——-

```
Number of digits = 4
gauravmathur@Annus-Mac Lab_03 % java reverse.java
Enter a number: 783
Reversed number = 387
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q11:Write a Java program to find sum of digits of a number.

Lab_03 > sum.java > ...

```java
package Lab_03;
import java.util.Scanner;

public class sum {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter a number: ");
        int num = sc.nextInt();
        int sum = 0;
        num = Math.abs(num);
        while (num != 0) {
        int digit = num % 10;
            sum = sum + digit;
            num = num / 10;
        }

        System.out.println("Sum of digits = " + sum);
        sc.close();
    }
}
```

OUTPUT:——-

```
Reversed number = 387
gauravmathur@Annus-Mac Lab_03 % JAVA SUM.JAVA
Enter a number: 45
Sum of digits = 9
gauravmathur@Annus-Mac Lab_03 %
```

2023BCSE008

Q:12:Write a Java program to keep taking input until user enters 0.

Lab_03 > input.java > ...

```java
package Lab_03;
import java.util.Scanner;
public class input {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the number:");
        int num;
        do {
            num = sc.nextInt();
        } while (num != 0);
        sc.close();
    }

}
```

OUTPUT:———

```
Sum of digits = 9
gauravmathur@Annus-Mac Lab_03 % JAVA INPUT.JAVA
Enter the number:
2
3
0
gauravmathur@Annus-Mac Lab_03 %
```

Ln 15, Col 1